SIH26143 — Deep Technical & Competitive Research

Satellite Oil-Spill Detection + AIS Vessel Attribution (NTRO)

EXECUTIVE VERDICT

Continue with SIH26143 — but only with modifications. The individual components (SAR oil-spill segmentation, drift/trajectory modelling, AIS anomaly detection) are each **mature, well-published research areas** with open datasets, open-source models, and even commercial operational services. Nothing you build will be the "first" system to combine SAR + AIS for spill attribution — that idea already exists in EU (EMSA CleanSeaNet), Norway (KSAT), and multiple 2023–2025 papers (see Part 9). **What does NOT exist as a public, reproducible, explainable, open-source pipeline** is the full chain — detection → age/geometry → backward drift reconstruction → AIS filtering → an auditable, uncertainty-aware ranking score — built specifically for the India/Indian Ocean context and demonstrable end-to-end in a hackathon timeframe. That integration + explainability + Indian-waters framing is your realistic innovation surface, not any single stage of the pipeline.

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1. Problem Understanding

The ask has three technically distinct sub-problems chained together: (a) computer vision on SAR/EO imagery, (b) physical oceanographic simulation (forward AND backward), (c) spatio-temporal data mining on AIS streams, unified by (d) an evidentiary scoring/ranking layer. NTRO's own framing ("estimate spill age," "trace backward," "rank potential vessels," never "identify the guilty vessel") signals they expect a probabilistic decision-support tool, not a legal-verdict engine — this framing should shape your whole product.

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2. Existing Research (Global)

PROVEN FACT — SAR oil-spill detection is a saturated research area. Dozens of papers 2019–2026 report CNN/U-Net/DeepLab/Transformer pipelines at 90–99% accuracy on the reference benchmark. Key line of work:

- **Krestenitis et al. 2019** (*Remote Sensing*, MDPI, doi:10.3390/rs11151762) created the field's standard benchmark: 1112 Sentinel-1 SAR images, 5-class segmentation (sea, oil spill, look-alike, ship, land), compared UNet/LinkNet/PSPNet/DeepLabv2/DeepLabv3+. <cite index="6-1">Instead of classifying certain dark formations as oil spills or look-alikes, this study applied semantic segmentation methods to categorize each pixel into sea surface, oil spill, look-alike, ship, or land classes, comparing UNet, LinkNet, PSPNet, DeepLabv2 and DeepLabv3+ across 1112 Sentinel-1 SAR images</cite>. This is >280 citations deep — treat it as the de-facto baseline you must at least match.
- Recent work (2024–2026) pushes toward **DeepLabv3+ region-specific retraining** for Suez Canal, achieving <cite index="26-1">98.14% accuracy, MIoU of 0.7872, and ROC area of 0.91 on Egyptian-specific data, versus 96.45% accuracy and 0.7161 MIoU for a generic EMSA-trained model</cite> — the finding that **region-specific fine-tuning beats a generic global model** is directly reusable as your justification for training/fine-tuning on Indian-Ocean-relevant scenes.
- **Look-alike discrimination remains the unsolved bottleneck.** <cite index="18-1">CNN, ResNet, Mask R-CNN, UNet, AlexNet, and Faster R-CNN have all been applied, but the issue of look-alikes persists and a mechanism to eliminate or minimize this is still needed</cite>. This is your legitimate, citable "unsolved problem" to target.
- Class imbalance is a documented, real limitation: <cite index="6-1">oil spills represent around 1.2% of total pixels in a typical scene, which hinders directly applying semantic segmentation to entire SAR scenes</cite>.
- 2026 work is moving toward **lightweight backbones for operational full-scene inference**: <cite index="2-1">a ResNet34-based model controls parameter size while maintaining spatial feature representation, enabling rapid inference on large-scale Sentinel-1 imagery and meeting operational-monitoring requirements, reflecting the mainstream accuracy-efficiency trade-off in remote-sensing detection</cite>.

- China's HY-1C/D programme has already run this at genuinely large scale: <cite index="3-1">a deep-learning framework was applied to more than 25,000 HY-1C/D CZI images between December 2018 and April 2023, revealing substantial oil spills in China's marginal seas, with spatial distribution and oil characteristics such as concentration, thickness and volume all quantified</cite>, and the same team showed the model transfers to Sentinel-1 SAR and to optical Landsat/Sentinel-2 imagery — i.e., **multi-sensor fusion at scale is already proven**, not speculative.

AIS + SAR attribution research (directly relevant, PROVEN to exist):

- **Ship tracing via bidirectional drift + AIS (2024, ScienceDirect)** is the closest published analogue to your entire proposed pipeline: <cite index="27-1">the method filters AIS data by the position of oil-spill areas on remote-sensing imagery, converts oil-spill areas into trajectory points, and then — based on a Lagrangian particle-motion model — proposes a bidirectional drift model that calculates the average similarity between forward and backward drift results to find the most likely responsible ship</cite>. **This paper essentially is SIH26143.** Read it in full before building anything; you must differentiate from it explicitly.
- **China joint SAR-AIS framework, Chang et al. 2024** (cited in a review): <cite index="23-1">this framework detects oil spills and identifies potentially responsible vessels, and the spatial/temporal correlation between oil slicks and vessel trajectories significantly improved source-attribution accuracy</cite>.
- A real enforcement case exists: <cite index="19-1">China proposed a framework combining SAR, AIS and boarding inspection for oil-spill accountability, applying it to track and identify a vessel illegally discharging oily sewage in the Bohai Sea — reportedly the first time China used SAR and AIS together in this way</cite>.
- Even without a spill event, dark/anomalous vessel patterns are already used by traders: <cite index="39-1">Kayrros's "Off-The-Radar Vessels Detection" uses satellite and AIS data to count vessels with AIS switched off worldwide, letting analysts correlate behavioural changes with sanctions and market events</cite> — proof that "AIS gap = suspicious behaviour" is an established, monetized signal, not a novel idea from your team.

Drift/trajectory modelling — solved, and freely usable as infrastructure, not a research target.

- **OpenDrift** (open source, MET Norway) natively supports what NTRO is asking for: <cite index="51-1">all instances of OpenDrift can be run in reverse — backwards from a final position — by reversing the sign of the advective increment, while nonlinear processes such as oil evaporation or vessel capsizing are disabled in backtracking mode; this simple backtracking scheme is a usable, much less complex alternative to iterative forward-modelling inverse methods</cite>. Its **OpenOil** module <cite index="48-1">is specifically designed for oil-spill drift and fate modelling, simulating oil weathering processes such as evaporation, emulsification, vertical mixing and biodegradation, with oil properties sourced from NOAA's ADIOS database</cite>.
- **NOAA GNOME** is the US government's own deployed tool: <cite index="52-1">GNOME is a desktop trajectory-modelling tool that works with downloadable Location Files containing pre-packaged tide and current data, alongside ADIOS, NOAA's stand-alone oil-weathering model</cite>.
- **Do NOT attempt to build a physics engine from scratch.** OpenDrift + OpenOil, wired to Copernicus Marine currents and ERA5/GFS winds, gives you research-grade forward AND backward drift for free, in Python, in days not months.

AIS anomaly/behaviour analysis — mature toolbox, mostly reusable off the shelf.

- Standard technique stack: <cite index="70-1">DBSCAN-based density clustering of trajectories, combined with dynamic time warping (DTW) for trajectory similarity, and kernel-density estimation for outlier scoring</cite> is the dominant paradigm across a decade of papers.
- More recent methods add Transformer-based trajectory prediction and SHAP-based explainability: <cite index="67-1">a BOA-XGBoost anomaly model achieved 98% and 93% detection accuracy on two real anomalous-behaviour types, with SHAP-based interpretability analysis confirming robustness</cite> — SHAP-style feature attribution is a good, easily citable technique for your explainable attribution score (Part 6).
- Classic risk-scoring already fuses spatial proximity with time: <cite index="68-1">DCPA (Dynamic Closest Point of Approach) and TCPA (Time to Closest Point of Approach) are standard maritime risk metrics combining spatial and temporal closeness</cite> — you can directly adapt DCPA/TCPA logic as two of your attribution-score features (distance-to-origin, time-to-spill).

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3. Indian Ecosystem

PROVEN FACT: India has an operational oil-spill drift-advisory service, but no public evidence it does AIS-based vessel attribution.

- INCOIS runs the **Online Oil Spill Advisory System (OOSA)**, now at Version 5.0: <cite index="30-1">INCOIS upgraded its Online Oil Spill Advisory System to Version 5.0 with advanced GIS and predictive capabilities, and provides oil-spill trajectory advisories on request, accessible via oosa.incois.gov.in</cite>. This is a **real-time operational government system for drift prediction** — it already solves the "predict future drift" requirement operationally in India. Your team should acknowledge OOSA explicitly rather than pretend it doesn't exist; your differentiator has to be backward attribution + AIS ranking, which OOSA does not appear to do publicly.
- INCOIS also runs the unrelated but relevant **SARAT (Search and Rescue Aid Tool)**, used recently for drift prediction of debris from the Wan Hai 503 fire off Kerala: <cite index="38-1">INCOIS activated its Search and Rescue Aid Tool to track possible drift patterns of containers, debris, or people who may have gone overboard, with simulations showing a 70–80% probability that drifting objects would move south-southeastward from the incident site over the next three days</cite>. This confirms India already has operational Lagrangian-drift infrastructure and expertise — a resource you can cite, not a competitor you need to beat on drift modelling alone.
- ISRO has flown actual detections: <cite index="31-1">ISRO's EOS-4 C-band SAR satellite was used for a preliminary detection of the Kerala oil spills following the capsizing of the Liberian container vessel MSC ELSA 3 near Alappuzha on 25 May 2025</cite>. So India has its own SAR asset for this exact use case — a strong "real Indian relevance" hook for your pitch, though you likely can't get EOS-4 raw tasking data as students (use Sentinel-1 instead, which is free).
- Bhuvan (NRSC/ISRO) has GIS-based spill mapping research going back over a decade — a "could not verify current operational status" item; treat any Bhuvan-based oil-spill layer as a prototype/legacy reference, not confirmed live capability.
- **Could not verify**: any NTRO-specific technology stack, since NTRO is India's classified technical-intelligence agency and does not publish its methods. Do not assume they have zero capability — assume the opposite, and that the SIH ask is precisely for the layer they *don't* already have automated (auditable AIS-ranking on top of drift + detection), which is why the PS explicitly separates "provide visual interface" as its own bullet — suggesting the ask is for a demonstrable decision-support prototype, not classified operational tooling.

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4. International / Commercial Ecosystem

System	Organization	Detection	SAR	AIS	Drift	Attribution	GIS	AI/ML	Access
CleanSeaNet	EMSA (EU)	Yes	Yes	Yes	Partial	Human-assisted	Yes	Semi-auto	Govt-restricted
Oil Spill Detection Service	KSAT (Norway)	Yes	Yes	Yes	Limited	Human-assisted	Yes	Semi-auto	Commercial
SAR + AIS	ICEYE	Yes	Yes	Yes	No	Manual correlation	Partial	Partial	Commercial
Off-the-Radar Vessels	Kayrros	No (vessel-focused)	Partial	Yes	No	N/A (sanctions, not spills)	Yes	Yes	Commercial
OOSA v5.0	INCOIS (India)	No	No	No	Yes	No	Yes	Predictive only	Public (govt)
GNOME/ADIOS	NOAA (US)	No	No	No	Yes	No	Limited	No	Free/open
OpenDrift/OpenOil	MET Norway	No	No	No	Yes (incl. backward)	No	No	Free/open	
BLOSOM	US (research)	No	No	No	Yes	No	No	No	Research

Key finding: **every operational system that does "attribution" (CleanSeaNet, KSAT, ICEYE) relies on a human analyst correlating AIS with the SAR detection** — <cite index="41-1">trained operators assess the images together with meteorological, oceanographic, and ancillary information such as AIS and vessel detection to identify possible pollution events and assist in identifying the source of the pollution</cite>, and <cite index="40-1">spills detected in SAR images are then correlated with AIS data to help pinpoint the source</cite>. **None of the commercial/government systems publish a fully automated, explainable, quantitative ranking algorithm for "most likely responsible vessel."** That is your real, defensible gap: automating what is currently a human-analyst judgment call, with an auditable score instead of tacit expertise.

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5. Dataset Audit

Satellite (SAR)

- **Krestenitis Oil Spill Dataset** — 1112 Sentinel-1 SAR images (~1250×650 px, VV band), 5-class masks (sea, oil, look-alike, ship, land). <cite index="10-1">It is the reference public benchmark for oil-spill detection in SAR imagery, with 280+ citations</cite>. Access: <cite index="10-1">distributed on request by the original authors (MKLab/CERTH), requiring a request from a supervisor's institutional email</cite> — plan for this lead time; request it in week 1.
- **Zenodo Sentinel-1 SAR Oil Spill dataset (Part I)**: <cite index="12-1">Sentinel-1 SAR images in Sigma0 (dB), 2048×2048 TIFF, with 1200 training/validation oil-spill images and matching binary ground-truth masks</cite>. URL: zenodo.org/records/8346860 — freely downloadable, no request process. Use this as your primary dataset if the Krestenitis request stalls.
- **DARTIS dataset (2025/2026, Eastern Mediterranean)**: <cite index="2-1">4,355 samples total (2,990 non-oil-spill, 1,365 oil-spill), 640×640 JPEG, covering coastal-topography and wind-field look-alike cases</cite> — good supplementary look-alike training data.
- **EMSA-CSN dataset**: <cite index="5-1">1100 labeled oil-spill incidents from the European Maritime Safety Agency's CleanSeaNet service</cite> — historically request-only, not confirmed to be open for students; treat as **could not verify** access.

AIS

- **MarineCadastre / NOAA AccessAIS** — the dataset actually named in the problem statement. <cite index="59-1">Since 2015 the data is distributed as CSV, filtered to one-minute intervals, in Zstd-compressed daily files covering all U.S. coastal waters, with MMSI unencrypted since 2015</cite>; also newer <cite index="58-1">2024/2025 GeoParquet-format monthly vessel-track files, analysis-ready and cleaned of sentinel-value anomalies</cite>. Fields include MMSI, lat/lon, SOG, COG/heading, vessel type, timestamp; <cite index="61-1">vessel_type/cargo codes follow the NMEA/Class A AIS specification, and since 2024 the AIS Vessel Identification Database (AVID) fuses 23 sources to correct or fill null vessel-characteristic fields</cite>. **Critical limitation: this is U.S.-coastal data only — it has zero Indian Ocean coverage.** Use it to prototype/validate your pipeline logic, but state clearly in your pitch that Indian-waters deployment would need Indian AIS (Coast Guard/DG Shipping) or a paid global AIS feed (e.g., MarineTraffic/Spire/exactEarth APIs), and fall back to **synthetic AIS** for the India-region demo — exactly as the official PS anticipates.
- **Global Fishing Watch** — free API, global AIS-derived fishing-vessel tracks; useful for realistic background traffic density even outside the US.
- Kaggle mirrors of MarineCadastre/AIS exist for quick prototyping without dealing with the raw CSV pipeline.

Oceanographic / Meteorological

- **Copernicus Marine Service (CMEMS)** — global ocean current, SST, wave reanalysis/forecast, free registration.
- **HYCOM** — global ocean circulation model, free.
- **ERA5 (ECMWF/Copernicus Climate Data Store)** — global reanalysis wind, pressure, precipitation, free, hourly, since 1940.
- All three plug directly into OpenDrift as forcing fields — <cite index="57-1">OpenOil allows coupling with coarser CMEMS, FVCOM, SHYFEM, CYCOFOS, HYCOM, and Norshelf forecasts for hydrodynamics, and NOAA, ECMWF and SKIRON wind fields via netCDF</cite> — so this is largely a data-plumbing task, not a modelling-research task.

Dataset verdict: Detection data is available (Zenodo mirror = no-request-needed fallback). AIS data for realistic India-region

demo is the weakest link — plan for synthetic AIS generation over Indian shipping lanes (seed near real ports: Kandla, JNPT, Mumbai, Kochi, Chennai, Vizag, Paradip) rather than assuming a real Indian AIS feed will be obtainable in hackathon time.

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6. AI/ML Methods for Detection

Model	Accuracy (lit.)	Cost	Data need	Explainability	Verdict for your team
U-Net (+ResNet backbone)	<cite index="15-1">Best published results correspond to U-Net with a ResNet backbone at 600 epochs</cite>	Low–med	Moderate	Medium	**Recommended baseline** — well-documented, small enough to train on Colab/Kaggle GPU
DeepLabv3+	<cite index="26-1">98.14% accuracy, 0.79 MIoU on region-specific data</cite>	Medium	Moderate	Medium	Good stretch goal if time allows
CNN classifier (dark-spot yes/no)	<cite index="4-1">99% classification accuracy with a 6-layer, 32-filter CNN</cite>	Very low	Low	High	Useful as a fast pre-filter before segmentation
Mask R-CNN / object detector	<cite index="7-1">AP of ~69% on Sentinel-1 object-detection benchmark</cite>	Medium-high	High	Low	Not worth the complexity for MVP
SegFormer / ViT-based	Emerging, few benchmarks yet	High	High	Low	Interesting "advanced" stretch, high risk

Recommendation: U-Net (ResNet encoder) for segmentation + a lightweight CNN pre-classifier to gate full-scene processing — this mirrors the "lightweight backbone + efficient inference" trend <cite index="2-1">reflected in the accuracy-efficiency trade-off that is the mainstream direction in current remote-sensing detection</cite>, while staying implementable by a student team in the available time.

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7. Drift Modelling Recommendation

Use **OpenDrift/OpenOil directly** rather than writing a custom Lagrangian solver. Justification: it is open-source, Python-native, already supports **backward-in-time seeding** (exactly what NTRO's requirement #3 asks for), already ingests CMEMS/HYCOM/ERA5, and already contains a NOAA oil-weathering database for evaporation/emulsification — building your own version of any of this would waste your limited hackathon time on solved infrastructure rather than the genuinely open attribution-scoring problem.

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8. AIS Methods Recommendation

Layered pipeline, each stage backed by literature:

1. **Geofence + time-window filter** around the drift-reconstructed origin (cheap, high-precision first pass).
2. **DBSCAN/trajectory clustering** to group and de-noise candidate tracks — <cite index="70-1">density-based clustering automatically classifies different vessel routes, a well-established technique across a decade of maritime-AIS literature</cite>.
3. **DTW-based trajectory similarity** between each candidate vessel's track and the reconstructed drift path, plus **DCPA/TCPA-style scoring** <cite index="68-1">combining spatial and temporal closest-approach metrics</cite>.

4. **Anomaly/behavioural flag** (speed drop, loitering, AIS gap) as a secondary signal, not a standalone verdict — Kayrros-style "AIS off = suspicious" logic <cite index="39-1">correlating AIS gaps with behavioural change</cite> is a good, citable precedent for this feature.

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9. Attribution Engine (Explainable Ranking)

PROVEN research support exists for using proximity + trajectory + temporal correlation as ranking features — this is not something you're inventing from nothing: <cite index="23-1">ships operating near an oil-spill location within a relevant time window can be flagged as potential sources, and abnormal vessel behaviour such as sudden speed reduction, prolonged drifting, unexpected route deviations, or repeated loitering may indicate illegal discharge, providing critical contextual validation beyond satellite imagery alone</cite>. The most directly transferable published scoring logic is the bidirectional-drift-similarity approach: <cite index="27-1">calculating the average similarity between forward and backward drift results to identify the most likely ship</cite>.

Do not invent arbitrary weights. Instead:

- Treat each factor (distance, time delta, trajectory-drift similarity, behavioural anomaly score, AIS continuity) as an independent, normalized sub-score (0–1).
- Combine with a **transparent, justifiable method** — e.g., a weighted sum where weights come from a small labeled/simulated validation set (via logistic regression or SHAP-derived feature importance, per <cite index="67-1">SHAP-based interpretability analysis used elsewhere in maritime anomaly research to confirm model robustness and interpretability</cite>), not hand-picked numbers.
- Report an explicit **confidence interval or uncertainty band** (Monte Carlo over drift-model uncertainty, since backward tracking accumulates position uncertainty over time) rather than a single deterministic score.
- Output label: **"Potentially associated vessel — rank N, confidence X%, evidence: [distance, time, trajectory match, anomaly flags]"** — never "guilty."

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10. Novelty / Research Gap

Already solved (well-published, don't re-research):

1. SAR oil-spill binary/multi-class segmentation (U-Net/DeepLab family).
2. Forward AND backward Lagrangian drift modelling (OpenDrift/GNOME).
3. AIS trajectory clustering and anomaly scoring (DBSCAN/DTW family).
4. SAR+AIS correlation as a *concept* for source identification (multiple 2023–2025 papers, multiple commercial services).
5. Region-specific fine-tuning beating generic global models.

Partially solved:

1. Look-alike vs. true-oil discrimination (persistent open problem).
2. Spill-age estimation (existing literature is thin; mostly inferred from oil-thickness/spreading models, not directly measured).
3. Fully automated (non-human-in-loop) end-to-end SAR→AIS attribution — exists in papers, not as public production software.
4. Uncertainty quantification on backward drift reconstruction.

Unsolved / genuinely open:

1. A public, open, explainable, auditable vessel-ranking *score* (as opposed to a binary "flag") with defensible feature weights.
2. Systematic handling of AIS spoofing/gaps as *adversarial* evasion in the attribution logic (most papers assume AIS is honest).
3. India/Indian-Ocean-specific tuning and demonstration (no public system does this).
4. Tight integration with a GIS decision-support interface aimed at non-technical maritime enforcement users.

Innovation ranking:

Innovation	Novelty	Feasibility	SIH Value	Research Value	Risk
Explainable, uncertainty-aware vessel ranking score	8	7	9	6	Med
Bidirectional drift + AIS filtering (India context)	5	8	8	4	Low
Look-alike discrimination improvement	7	4	5	8	High
AIS-spoofing/gap-aware attribution	8	5	7	7	High
Full GIS dashboard for enforcement users	4	9	8	3	Low

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11. Proposed Architecture — Stage-by-Stage Reality Check

Stage	Already common?	Feasible in SIH time?	What to actually build	What to skip
SAR ingestion	Yes	Yes	Sentinel-1 via Copernicus Data Space API	Custom satellite tasking
Detection/segmentation	Yes (saturated)	Yes	Fine-tune U-Net on Zenodo/Krestenitis data	Training a new architecture from scratch
Spill geometry	Yes	Yes	Simple polygon/area/length from mask	—
Spill age estimation	Rare/thin literature	Partially	Heuristic from area+shape+known weathering curves; flag as "estimate," not precise science	Do not claim precise age — this is scientifically shaky
Ocean/weather ingestion	Yes	Yes	CMEMS + ERA5 API pulls	Building your own ocean model
Backward drift	Yes (OpenDrift)	Yes	Wire OpenDrift's native backtracking	Custom Lagrangian solver
AIS historical search	Yes	Yes (with synthetic data caveat)	Geofence + time-window query	Real-time global AIS ingestion (licensing cost)
Vessel filtering	Yes	Yes	Rule-based + DBSCAN	—
Trajectory correlation	Yes	Yes	DTW / DCPA-TCPA	—
Behaviour analysis	Yes	Yes, lightweight version	Speed/heading anomaly flags	Deep Transformer trajectory-prediction model (overkill)
Attribution scoring	Partially open	Yes	Weighted, explainable, uncertainty-banded score	Black-box ML classifier with no explanation
GIS dashboard	Yes	Yes	Leaflet/Mapbox + your own API	Building a full enterprise GIS suite

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12. MVP vs Advanced Scope

MUST HAVE

- SAR upload/fetch → U-Net segmentation → spill polygon + area.

- OpenDrift backward-drift reconstruction from spill polygon using real (or cached) CMEMS/ERA5 data.
- AIS ingestion (MarineCadastre real data for demo validity + synthetic data layered over an Indian coastal scenario) with geofence+time filtering.
- Explainable ranking score (distance, time, trajectory similarity) with a clear "potentially associated" output.
- Web GIS map showing spill, drift cone, and ranked candidate vessels with rationale.

SHOULD HAVE

- Spill-age heuristic estimate.
- Basic AIS anomaly flags (speed drop, course deviation).
- Confidence bands on the drift cone (Monte Carlo).

NICE TO HAVE

- Look-alike vs. oil discrimination improvement.
- Multi-satellite fusion (optical + SAR).
- AIS-gap-aware scoring for evasion detection.

DO NOT BUILD

- A custom oceanographic/physics model.
- A legally worded "guilty vessel" determination.
- Real-time global satellite tasking (you don't have access).
- A full production-grade enterprise GIS platform.

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13. Failure Modes

Failure mode	Impact	Mitigation	Confidence handling
SAR look-alikes (biogenic films, low wind)	False positive spills	Multi-feature classifier + wind-speed gating (5–18 m/s window per literature)	Flag as "low confidence — possible look-alike"
Cloud cover (optical only)	Missed detections	Prioritize SAR (all-weather) as primary sensor	N/A
Missing/sparse AIS near origin	No attribution possible	Expand search radius/time window, flag "insufficient AIS coverage"	Explicit "cannot attribute" output
AIS spoofing/gaps	Wrong or missed vessel	Treat gaps as a scored anomaly feature, not silent failure	Lower confidence, surface the gap explicitly
Inaccurate ocean current/wind forcing	Wrong backward-drift origin	Use ensemble/multiple forcing sources, report drift-cone uncertainty	Uncertainty band on map
Multiple vessels near origin	Ambiguous attribution	Rank list, not single answer	Show top-N with scores
Old spill / natural seep	False attribution attempt	Age heuristic + seep-location cross-reference (known seep zones)	Flag "possible natural seepage"
Rough sea state	Reduced SAR sensitivity	Note environmental caveat in output	Lower detection confidence

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14. SIH Evaluator Assessment

Criterion	Score /10
Problem relevance	9
Innovation	6
Technical complexity	8
AI/ML depth	7
Data engineering	8
GIS capability	7
Real-world usefulness	8
Government usefulness	8
Demonstrability	7
Scalability	6
Dataset feasibility	6 (India-specific AIS is the weak spot)
Prototype feasibility	7
Explainability	8 (if you build it right)
Reliability	6
Novelty	6
Presentation potential	8

Top 5 reasons you could win:

1. Genuine multi-domain technical depth (CV + oceanography + data mining) that most teams won't fully implement.
2. Explainability/auditability angle directly matches how NTRO/judges will want a "non-accusatory" tool.
3. Strong open-source scaffolding (OpenDrift, U-Net, DBSCAN) means you can spend your limited time on integration + polish, not reinventing wheels.
4. Compelling, visual GIS demo (drift cone + ranked ship markers) is inherently demo-friendly.
5. Real Indian relevance (Kerala 2025 spills, INCOIS OOSA) gives you a strong narrative hook.

Top 5 reasons you could lose:

1. No real Indian AIS data — judges may probe this and call the demo unrealistic if you don't own the limitation upfront.
2. Spill-age estimation is scientifically weak territory; overclaiming precision here will get you picked apart.
3. Teams that don't manage scope will try to build a physics engine from scratch and run out of time.
4. If your "attribution" reads as accusatory rather than probabilistic, evaluators (and legally-minded judges) may push back hard.
5. Judges familiar with EMSA/KSAT/ICEYE may say "this already exists commercially" — you must proactively address this in your pitch (see Part 4).

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15. Final Recommendation

1. Continue with SIH26143? YES, WITH MODIFICATIONS — scope tightly around the ranking/explainability layer, not novel detection or drift science.

2. Genuine innovation potential? YES — in the explainable, uncertainty-aware, India-context integration layer; NOT in any individual component.

3. Technically feasible for a student team? PARTIALLY — feasible if you use existing open-source infra (OpenDrift, pretrained/fine-tuned U-Net, DBSCAN) rather than building each layer from scratch; not feasible if you try to out-research published papers on detection or drift modelling.

4. Central innovation: An **explainable, uncertainty-quantified vessel-attribution scoring engine** that turns today's human-analyst SAR+AIS correlation (as done at EMSA/KSAT) into an auditable, reproducible algorithm — with an explicit "cannot attribute" / confidence-banded output instead of a false-certainty verdict.

5. Final product description: A web-based GIS decision-support tool where an analyst uploads/selects a SAR scene, the system segments and characterizes the spill, reconstructs its probable drift origin backward in time using real oceanographic/weather forcing, cross-references historical AIS traffic near that origin, and returns a ranked, explainable, confidence-scored shortlist of "potentially associated vessels" — never a legal verdict.

6. Avoid building: custom physics/ocean models, a from-scratch drift solver, deep Transformer trajectory-prediction models, any "guilty vessel" framing, real-time global satellite tasking claims.

7. Three strongest research gaps: (a) no public automated+explainable attribution scoring engine exists — only human-analyst correlation; (b) no India/Indian-Ocean-specific instantiation of this pipeline exists publicly; (c) AIS-gap/spoofing-aware attribution is barely addressed in the literature.

8. Three strongest differentiators: (a) explainability + uncertainty quantification instead of a black-box score; (b) end-to-end integration built on proven open-source components rather than one narrow research contribution; (c) explicit "non-accusatory," legally-careful framing matching NTRO's own wording.

9. Recommended final architecture: Sentinel-1 → U-Net segmentation → geometry/age heuristic → CMEMS/ERA5-forced OpenDrift backward reconstruction with uncertainty ensemble → geofenced historical AIS query (real MarineCadastre data + synthetic Indian-waters overlay) → DBSCAN/DTW/DCPA-TCPA feature extraction → transparent weighted-score ranking with confidence bands → web GIS dashboard.

10. Final SIH strategy (10 bullets):

1. Lead your pitch with "we studied EMSA/KSAT/ICEYE — here's exactly what they do and what they don't automate."
2. Get the Zenodo SAR dataset immediately; request the Krestenitis dataset in parallel as backup/supplement.
3. Wire OpenDrift + CMEMS + ERA5 in week one — this is infrastructure, not your differentiator, so don't let it eat your timeline.
4. Build the explainable scoring engine first, since it's your actual innovation; detection and drift can be "good enough."
5. Use MarineCadastre AIS for validating pipeline logic, and clearly-labeled synthetic AIS for the India-region narrative demo.
6. Never claim a vessel is "guilty" — always "potentially associated, confidence X%."
7. Show uncertainty visually (a drift cone, not a single line) — judges notice scientific honesty.
8. Prepare a one-slide "failure modes we handled" (look-alikes, AIS gaps, natural seeps) — this signals technical maturity.
9. Explicitly cite INCOIS OOSA and Kerala 2025 EOS-4 detection as India-context grounding, not as competitors to dismiss.
10. Rehearse a crisp answer to "why hasn't NTRO built this already" — your answer: they likely have the classified pieces, but not a demonstrable, explainable, publicly presentable civilian-facing decision-support layer.

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16. Sources

Papers

- Krestenitis, M. et al. (2019). *Oil Spill Identification from Satellite Images Using Deep Neural Networks*. Remote Sensing 11(15), 1762. https://doi.org/10.3390/rs11151762

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Government/Operational

- INCOIS Annual Report 2023–24 (OOSA v5.0). https://www.incois.gov.in/documents/Reports/AnnualReports/AR_2023-24_20250813165305.pdf
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- KSAT Oil Spill Detection Service. https://www.ksat.no/earth-observation/environmental-monitoring/oil-spill-detection-service/
- ICEYE blog, Turku Archipelago spill case. https://www.iceye.com/blog/timely-sar-data-speeds-up-marine-oil-spill-response-two-example-cases
- Kayrros Off-The-Radar Vessels Detection. https://www.kayrros.com/products/off-the-radar-detection-for-traders/

Datasets

- Zenodo: Sentinel-1 SAR Oil Spill dataset, Part I. https://zenodo.org/records/8346860
- MarineCadastre / NOAA AccessAIS. https://hub.marinecadastre.gov/datasets/vessel-traffic-ais-1 ; https://marinecadastre.gov/ais/
- GitHub ocm-marinecadastre/ais-vessel-traffic. https://github.com/ocm-marinecadastre/ais-vessel-traffic

SIH Context

- SIH 2026 Master Problem Statement Catalogue (confirms SIH26143 as newly listed for 2026, NTRO). https://www.blinknbuild.in/Assets/SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf

THE ONE-SENTENCE STRATEGY

Build not another oil-spill detector or drift model — both are solved and freely available — but the missing **explainable, uncertainty-aware attribution engine** that turns the manual SAR+AIS correlation currently done by human analysts at EMSA/KSAT into an auditable, reproducible, India-context-ready scoring algorithm that ranks "potentially associated" vessels without ever claiming certainty it hasn't earned.